

4.4 Managing the impacts of tectonic hazards

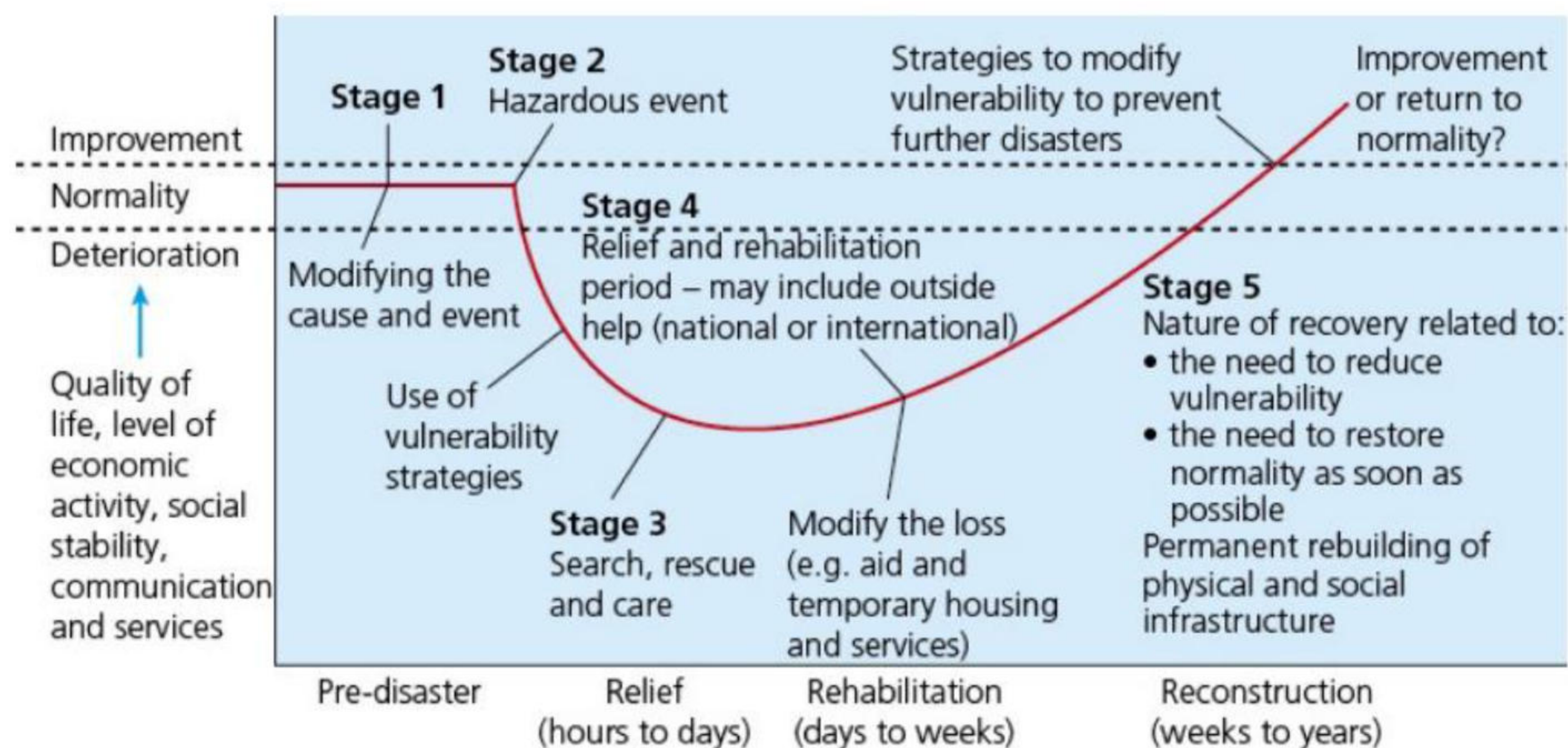
This chapter will explain:

- ★ the primary and secondary responses to tectonic hazards
- ★ the strategies and techniques used to manage the impacts of tectonic events
- ★ detailed specific example: Tiburon Peninsula earthquake, Haiti, 2021
- ★ detailed specific example: Mount Etna, Sicily, Italy.

The primary and secondary responses to tectonic hazards

The Park Model shows the primary and secondary responses to natural hazards ([Figure 4.18](#)). The primary responses to tectonic hazards cover the short-term responses, that is search, rescue and relief, while the secondary responses refer to the long-term activities, such as rehabilitation and reconstruction.

- » Primary responses are those that occur in the first few days or weeks following a natural disaster. They include attempts to recover bodies, search and rescue, medical care for those with injuries, accommodation for those whose homes have been damaged or destroyed, and the provision of basic amenities such as food, clean water, shelter and healthcare.
- » Secondary responses usually occur after the first few weeks or months and can continue for years. They include the rebuilding of housing, education, and improving infrastructure. In many LICs, reconstruction may take decades.



▲ **Figure 4.18** The primary and secondary responses to hazards – the Park Model

Activities

- 1 Distinguish between primary and secondary responses.
- 2 Identify whether the following are primary or secondary responses:
 - a Rebuilding homes
 - b Providing tents
 - c Providing toilets
 - d Rebuilding bridges.

Managing the impacts of tectonic events

The impacts of volcanic eruptions can be reduced in several ways:

- » Spraying lava flows with water to cool them down, helping them to solidify. This was successfully carried out in Heimaey, Iceland. The islanders used sea water to spray on to the lava flow and managed to divert it away from reaching the town. In contrast, some residents of Goma in the Democratic Republic of the Congo tried to spray water on to lava flows from the Nyiragongo volcano, but this proved unsuccessful as they had neither enough water nor the infrastructure to deal with the advancing lava flows.
- » Digging diversion channels to divert lava flows away from settlements: This has been used successfully

on Mount Etna, Sicily, although the lava needs to be diverted away from human habitation and infrastructure.

- » Adding 'cold' boulders to a lava flow in an attempt to cool the lava and stop it moving.

However, if the eruption is a pyroclastic flow, there is little that can be done to prevent the impacts other than evacuation.

Predicting volcanoes

The main methods of predicting volcanoes include:

- » using seismometers to record swarms of tiny earthquakes that occur as the magma rises
- » using chemical sensors to measure increased sulphur levels
- » using lasers to detect the physical swelling of the volcano
- » measuring small-scale uplift or subsidence, changes in rock stress and changes in radon gas concentration
- » using ultrasound to monitor low-frequency waves in the magma resulting from the surge of gas and molten rock, as happened with Mount Pinatubo, El Chichón and Mount St. Helens.

Overall, relatively few people are killed by volcanic eruptions. This may be because most of the world's active volcanoes are monitored, especially in areas with high population density. Nevertheless, volcanoes can erupt without warning. If Vesuvius in Italy were to erupt again, millions of people could be affected. In addition, high-magnitude events, such as a VEI 8 supervolcano, will occur again at some point, and there is little that can be done to reduce some of the global impacts of such an event.

Activities

- 1 Outline two ways of controlling lava flows.
 - 2 Outline two ways in which volcanoes can be predicted.
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Preparing for earthquakes

The main ways of preparing for and managing earthquakes include:

- » better forecasting, warning, evacuation and emergency procedures
- » improving building design and safe houses
- » land-use planning for building location.

Forecasting and warning

There are a number of ways of predicting and monitoring earthquakes. These include:

- » measuring crustal movement – the small-scale movement of plates
- » recording changes in electrical conductivity
- » noting strange and unusual animal behaviour, for example among fish (such as carp) or toads
- » checking historical evidence – there are possibly trends in the timing of earthquakes in some regions.

Evacuation and emergency procedures

Loss of life can be minimised by ensuring that the population is educated in emergency evacuation procedures. For example, schools in Japan practise earthquake drills, and there is disaster education in many schools, colleges and states in the USA.

The US Centers for Disease Control and Prevention (CDC) provides information on preparing for hazards, for example 'drop, cover, and hold on' in the case of an earthquake. In an earthquake, people should get outside if possible. If trapped indoors, they should shelter under a sturdy object such as a kitchen table. When the shaking has stopped, they should get outside and move away from the building.

Building design

Buildings can be designed to cope with the shock waves that occur in an earthquake. For example, single-storey buildings are more able to absorb shock waves than multi-storey buildings as the potential for swaying is reduced. Also, tall buildings can be built with a 'soft storey' at the bottom, such as a car park on raised pillars. This soft storey may collapse in an earthquake so that the upper floors sink down on to it, which cushions the impact.

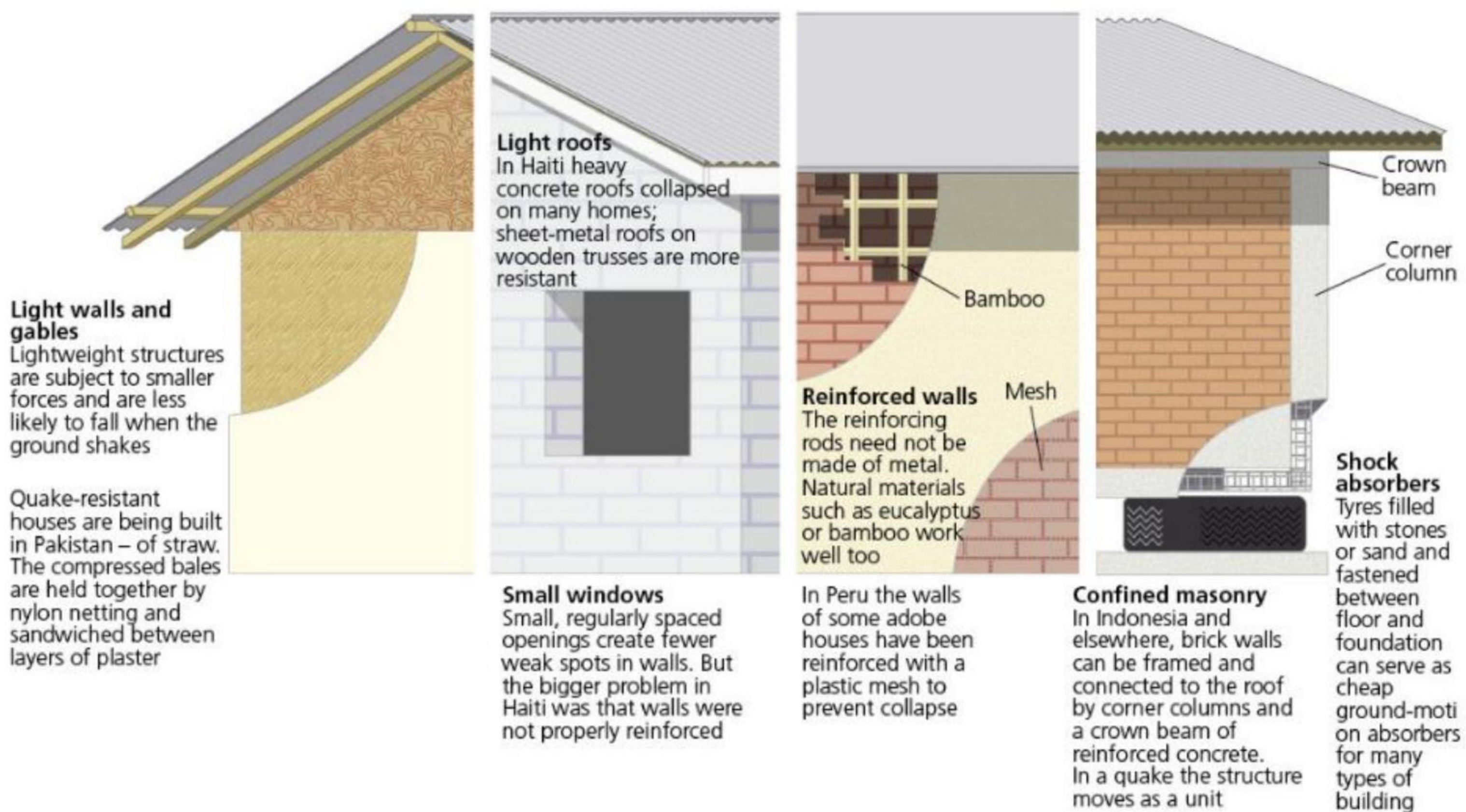
Building reinforcement strategies include building on foundations that are laid deep into the underlying bedrock, and steel frames that can withstand shaking can also be used.

In an earthquake, high-rise buildings respond slowly, and shock waves are increased as they move up the building. If buildings are too close together, vibrations may be amplified between buildings and increase the damage. Therefore maximising space between buildings can help minimise this amplification.

Safe houses

Poverty, corruption and poor governance mean that buildings in many countries are not earthquake-proof. Billions of people live in houses that cannot withstand shaking. Yet safer ones can be built cheaply, using straw, adobe (cheap, sun-dried brick) or old tyres, by applying a few general principles ([Figure 4.19](#)).

In rich cities in fault zones, the added expense of making buildings earthquake-resistant has become a fact of life. Concrete walls are reinforced with steel, for instance, and a few buildings even rest on elaborate shock absorbers. Strict building codes were credited with saving thousands of lives when a magnitude 8.8 earthquake hit Chile in February 2010. But in less developed countries, like Haiti, conventional earthquake engineering is often unaffordable, even though there are some cheap solutions.



▲ **Figure 4.19** Safe-house design

In Peru in 1970 an earthquake killed more than 70,000 people, many of whom died when their houses crumbled around them. Heavy, brittle walls of traditional adobe cracked instantly when the ground started to move. Existing adobe walls can be reinforced with a strong plastic mesh installed under plaster. During an earthquake, these walls crack but do not collapse, allowing occupants to escape. Plastic mesh could also work as a reinforcement for concrete walls in Haiti and elsewhere.

Researchers in India have successfully tested a concrete house reinforced with bamboo. A model house for Indonesia rests on ground-motion dampers – old tyres filled with bags of sand. Such a house might be only a third as strong as one built on more sophisticated shock absorbers, but it would also cost much less – and so is more likely to get built in Indonesia. In northern Pakistan, straw is available. Traditional houses are built of stone and mud, but straw is far more resilient, and warmer in winter.

Building location

Land-use planning is another way of reducing earthquake risk. Densely populated areas and important services, such as hospitals and fire services, should not be built close to known fault lines or areas prone to landslides. Areas with weak rocks, faulted (broken) rock and soils should, ideally, be avoided for buildings. In California, since 1972, no building can be built within 17 m of a surface fault line, although this would not offer much protection from an earthquake.

Ultimately, although it is possible to suggest areas where most earthquakes are likely to occur, it is not possible to predict with certainty exactly where they will occur, when they will occur or their magnitude. However, taking precautions may help reduce their impact.

Certain areas are very much at risk from earthquake damage – notably areas with weak rocks, faulted rocks and soft soils. Many oil and water pipelines in tectonically active areas are built on rollers so that they can move with an earthquake rather than fracture ([Figure 4.20](#)).

Activities

- 1 Identify ways in which earthquakes can be predicted.
- 2 Explain the term 'safe house'. Briefly explain how houses can be made 'safe'.



▲ **Figure 4.20** A pipeline on rollers to allow movement during an earthquake

Detailed specific example

Tiburon Peninsula earthquake, Haiti, 2021

Haiti is located on the Gonâve Microplate, a small strip of the Earth's crust that is squeezed between the North American and Caribbean Plates. This makes it vulnerable to earthquakes. In August 2021, a 7.2 magnitude earthquake struck the country's Tiburon Peninsula. It was a shallow-focus earthquake (10 km deep) and the epicentre was about 150 km west of the capital Port-au-Prince ([Figure 4.21](#)). The earthquake was caused by the eastward movement of the Caribbean plate relative to the North American plate.

The earthquake caused around 2250 deaths, more than 12,750 injuries and damage estimated at US\$1.5–1.7 billion. There were over 900 aftershocks, two of which were in 2022, killing a further two people. The earthquake was followed by the impact of Hurricane Grace the next day, which triggered thousands of landslides. At least 137,500 buildings were damaged or destroyed. Les Cayes, Haiti's third-largest city, was badly affected, with over 37,000 homes destroyed and 46,000 others damaged, see [Figure 4.22](#). More than 50 medical facilities were damaged and six destroyed, and over 1000 schools were damaged or destroyed.

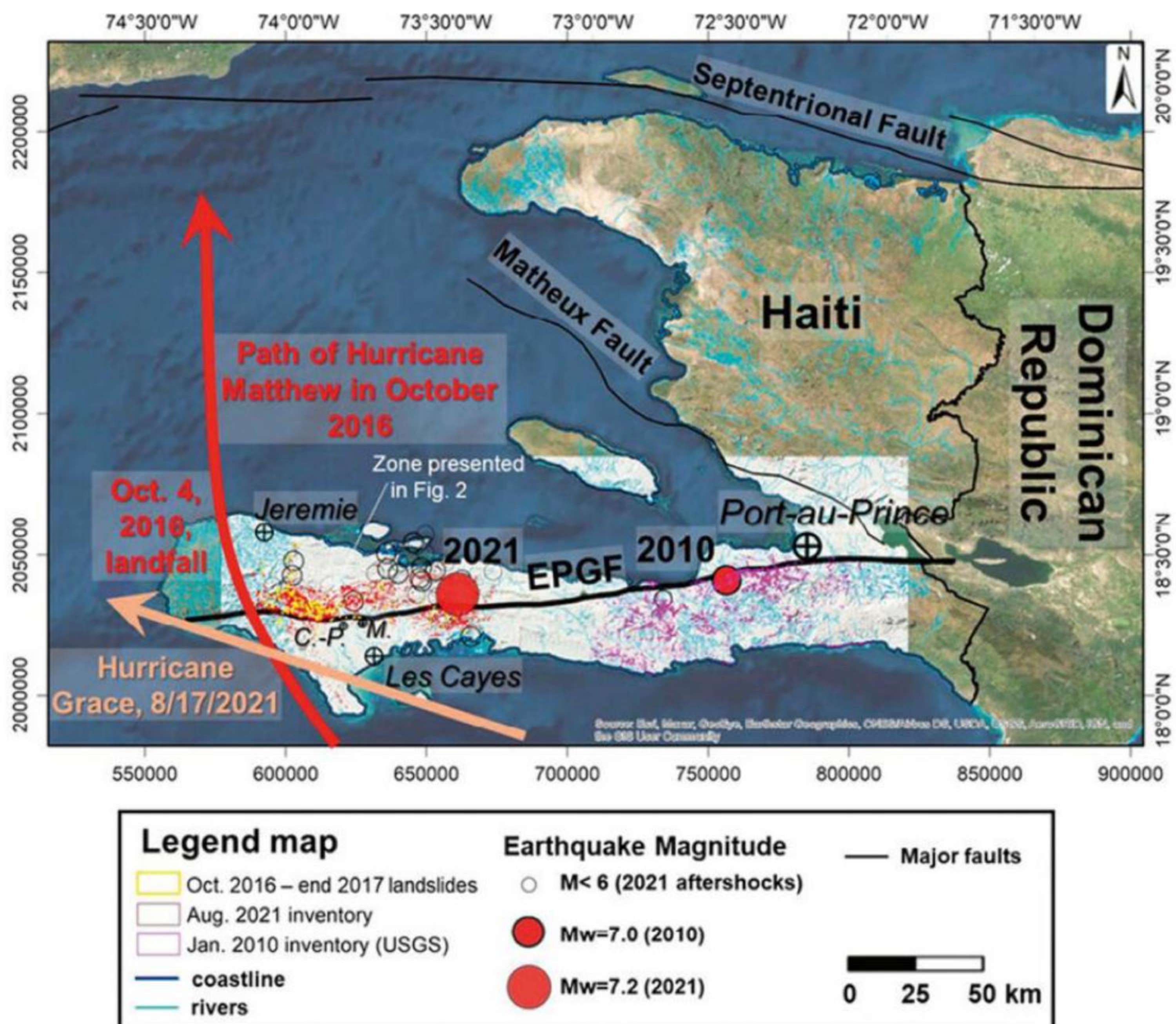
Shelter was the greatest long-term need. Emergency services were required given the needs of the population – including food, water, medicines and Covid-19 vaccines. Rebuilding is ongoing, including to infrastructure (communications, transport, WASH (water, sanitation and hygiene) and electrical supplies).

Six months after the earthquake, Haiti began to look at long-term recovery and reconstruction. It needed around US\$2 billion to rebuild social services and housing, health and education infrastructure, and food supply. After that, aid funds were spent on agriculture, industry and key infrastructure. Haiti's financial situation meant it was reliant on international aid for its recovery. It is the lowest-income country in the Western Hemisphere and is included by the UN on its list of the 45 least developed countries in the world, the only country from the Caribbean.

However, many donor countries were facing their own financial problems or were less able or unwilling to provide aid. In addition, Haiti was competing with other countries for funding, notably Afghanistan, Ethiopia, Ukraine, and Turkey and Syria (following the earthquake in 2023).

Research using remote sensing of the damage to buildings found that those with deep foundations and timber roofs fared best during the earthquake. The research, undertaken by international scientists with the aid of local volunteers who took notes and uploaded photographs, has helped ensure that new buildings will be more earthquake-proof for the future.

According to the World Bank, the 2021 Disaster Risk Management and Reconstruction (DRM) Project for Haiti focused on supporting Haiti to improve its disaster-response capacity, and enhancing the resilience of critical transport infrastructure.



▲ **Figure 4.21** The location of the Tiburon Peninsula earthquake and the track of Hurricane Grace

Key activities and systems developed under the project included risk assessments, communications and early warning systems, and improved road sections. In response to the August 2021 earthquake, the government was able to:

- mobilise Municipal Civil Protection Committees
- conduct a rapid and sound assessment of damages and losses
- achieve greater coordination among groups and individuals working in the DRM system.

Activities

- 1 Explain why earthquakes occur in Haiti.
- 2 Suggest how government actions after the 2021 earthquake helped manage the recovery process.



▲ **Figure 4.22** Search and rescue can be physically demanding as well as extremely dangerous

Mount Etna, Sicily, Italy

Mount Etna is an active stratovolcano on the east coast of the island of Sicily, Italy. It is located on the convergent plate margin of the African and Eurasian Plates (Figure 4.23). It is one of the world's most active volcanoes. In 2021 it increased in height by approximately 30 m, to around 3369 m. In historic times, only 77 deaths have been caused by its eruptions, the last two being in 1987. Nevertheless, Etna is a very active volcano.

There are different types of eruption at Etna. Most occur at one of the five craters near the summit (Figure 4.24). There are about 300 side vents, ranging in size from large craters to small holes. Ironically, the highly explosive summit eruptions are less hazardous than the flank eruptions, as these can occur close to inhabited areas. Since 2000 there have been at least four periods of flank eruptions and six periods of summit eruptions.

Volcanic activity has occurred at Etna for around 500,000 years. It evolved in three main phases from a shield volcano caused by submarine fissure eruptions to a stratovolcano caused by explosive activity. Over time, volcanic activity has moved from the southeast to the northwest.



▲ **Figure 4.23** The location of Mount Etna and tectonic plates



▲ **Figure 4.24** The craters near the summit of Etna

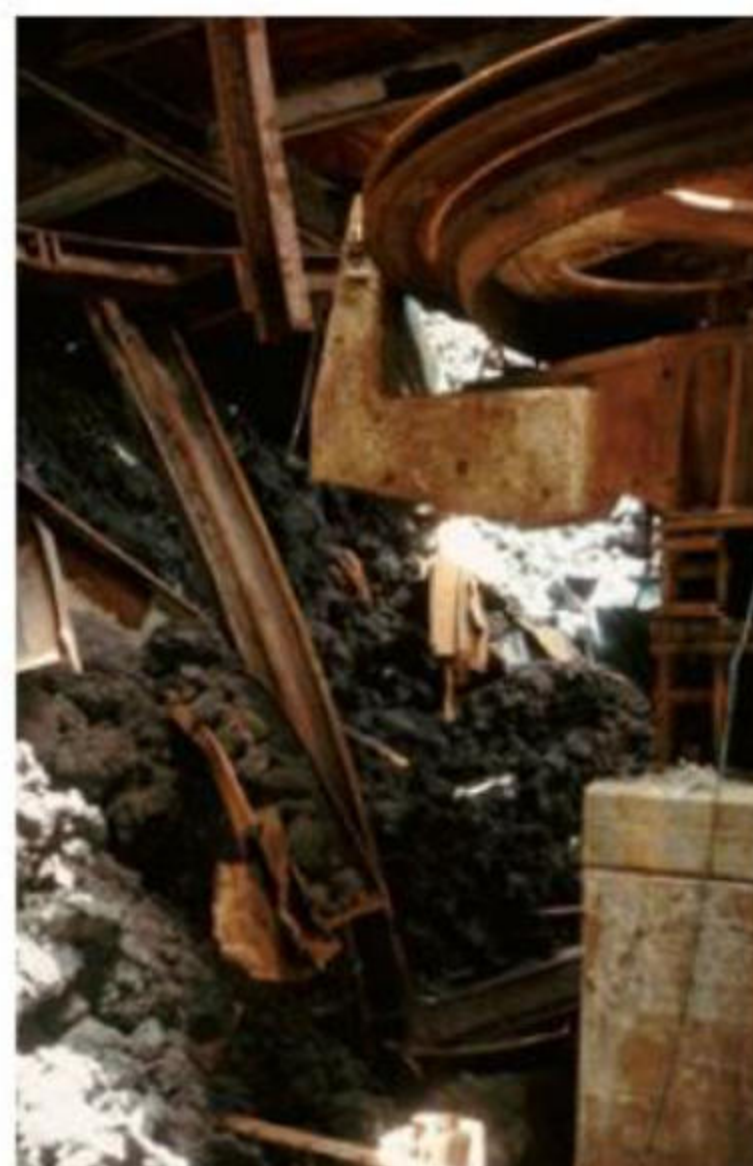
Around 8000 years ago, a massive landslide caused a large depression on the side of Etna – the Valley of the Ox – and triggered a large tsunami. Major eruptions have caused the summit to collapse many times, forming calderas. A collapse 2000 years ago formed the Piano Caldera, although this has largely been filled in by subsequent lava flows.

In 1971 lava flows buried the Etna Observatory and destroyed the Etna cable car. The 1991–93 eruptions threatened the town of Zafferana, but diversion of the lava flow saved the town. Initially the lava flow overtopped the barriers, so engineers used explosives near the source of the lava flow to divert it away from Zafferana.

Between 1995 and 2001 there was an increase in volcanic activity. In 2001 an eruption on the south side of Etna occurred during the tourist season and damaged the cable car station ([Figure 4.25](#)). In 2002–03, a large eruption destroyed the tourist station Piano Provenzana. Some of the footage of this eruption was used in the film *Star Wars: Episode III – Return of the Sith*.

Some eruptions continue for a very long time. One in 2008, for instance, lasted for 417 days. Eruptions in 2011 and 2012 led to the closure of Catania Airport several times. Starting in 2021, a series of explosive eruptions affected surrounding settlements, with rocks and volcanic ash falling as far away as Catania. Etna erupted 11 times in just three weeks.

The Global Volcanism Program reports that since 1955 Etna has experienced 17 VEI 1 eruptions, 24 VEI 2 eruptions and seven VEI 3 eruptions.



▲ **Figure 4.25** Damage to tourist infrastructure by lava flows at Etna, Sapienza Refuge ski area

There have been many responses to the threat of volcanic activity on Mount Etna. In the 1970s the government took photos of Etna temperature readings in order to monitor change. During an eruption in 1992 they built an earth barrier that was 200 m long and 20 m high to prevent lava from reaching Zafferana. It contained the lava flow for three months. US marines used explosives to blast a hole in the lava, and then dropped blocks of concrete into it to cool it down. During an eruption in 2002 evacuations occurred, schools were closed and a state of emergency was declared. The Institute of Vulcanology monitors radon gas, uses GPS to examine any changes in size or shape of the volcano, and uses very sensitive seismometers to detect earth tremors. Following the eruptions of 2002–3 the government pledged financial aid to farmers and those working in the tourism industry.

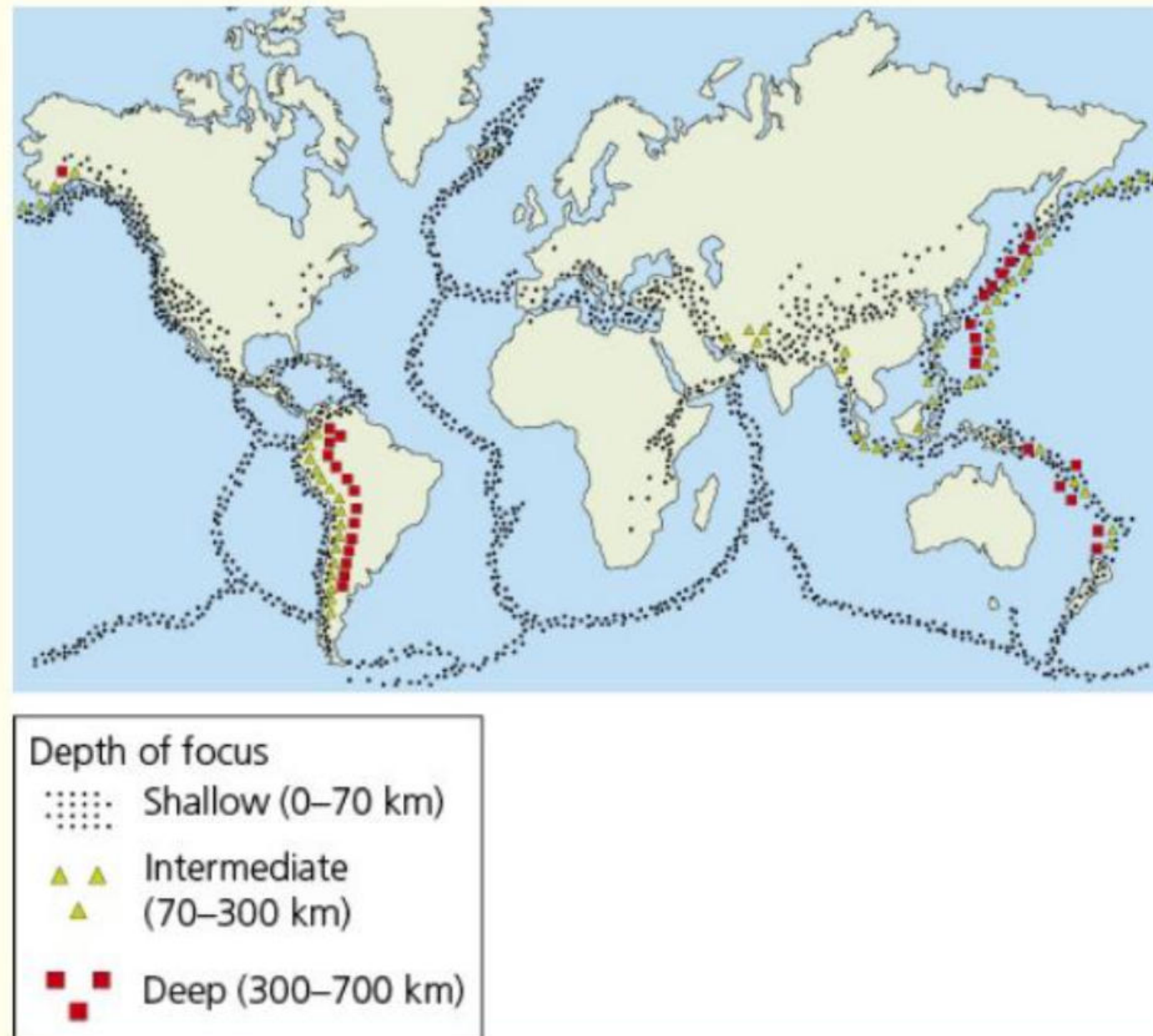
Despite the relatively low number of fatalities, Etna has proved disruptive to many local residents and to the tourism sector. Methods used to control the impacts of Mount Etna have included constructing earthen dams to divert lava, bombing the lava flows, cooling lava using sea water, blocking lava tunnels to reduce the lava flow and monitoring air quality (for sulphur dioxide) and ground deformation to make predictions about likely activity.

Activities

- 1 Explain the impacts of volcanic activity on Mount Etna.
- 2 Analyse ways in which the threats posed by Mount Etna have been managed.

Practice questions

- 1 [Figure 4.26](#) shows the distribution of shallow-, intermediate- and deep-focus earthquakes.



▲ **Figure 4.26** The distribution of shallow-, intermediate- and deep-focus earthquakes

- a Describe the distribution of shallow-focus and deep-focus earthquakes. [4]
 - b Suggest reasons to explain the distribution of shallow-focus and deep-focus earthquakes. [4]
- 2 [Figure 4.27](#) shows the summit of Mount Etna in Sicily.



▲ **Figure 4.27** The summit of Mount Etna

- a Identify **two** features of the volcano shown on the photograph. [2]
 - b Describe **two** hazards related to volcanoes. [4]
 - c Explain **two** reasons why people live in areas at risk from volcanic activity. [4]
- 3 Assess the extent to which the impacts of earthquakes can be managed. [7]